

ESP32-S3 1.28" Star Wars Tactical Radar Setup Guide

This guide explains how to set up and flash the Star Wars Tactical Radar display onto the Waveshare ESP32-S3 Touch LCD 1.28 board using Arduino IDE.

GitHub Repository

Project Files (.ino and Aurebesh8pt7b.h):

<https://github.com/bjwest9738/ESP32-1.28-Star-Wars-Tactical-Radar>

Hardware Used

- Waveshare ESP32-S3 Touch LCD 1.28
- USB-C data cable (not power-only)
- Normally Open reed switch (optional)
- Magnet mounted to servo horn (optional)

Required Libraries

Library Name
Adafruit GFX Library
Adafruit GC9A01A
Adafruit BusIO

Install Libraries

Arduino IDE → Sketch → Include Library → Manage Libraries

Search and install each required library listed above.

Required ESP32 Board Package

Install 'esp32 by Espressif Systems' through Arduino IDE Boards Manager.

Recommended Arduino IDE Settings

Setting	Value
Board	ESP32S3 Dev Module
Upload Speed	921600
USB CDC On Boot	Enabled
Serial Monitor	115200 baud

Important Font File

The file named 'Aurebesh8pt7b.h' must stay in the same folder as the .ino sketch file.

Waveshare Display Pins

Function	GPIO
Backlight	GPIO2
DC	GPIO8
CS	GPIO9
CLK	GPIO10
MOSI	GPIO11
MISO	GPIO12
RST	GPIO14
Reed Switch	GPIO15

Reed Switch Wiring

GPIO15  Reed Switch  GND

Use a Normally Open reed switch. No external 3.3V wire is needed.

Common Problems

- No COM port visible: usually caused by a power-only USB cable
- Adafruit_GFX.h missing: install required libraries
- Black screen: check TFT_BL backlight code
- Upload hangs: hold BOOT while uploading

Created for the ESP32-S3 Waveshare 1.28 Tactical Radar project.